

DataMind AI

Machine Learning Analysis Report

Generated On : 02 July 2026 22:52

This report contains the complete analysis performed by DataMind AI.

Dataset Summary

Property	Value
Problem Type	classification
Target Column	accident
Recommended Model	Logistic Regression
Confidence	85

Model Training Results

Rank	Model	Score
1	Random Forest	0.7449
2	Logistic Regression	0.7372
3	XGBoost	0.7346
4	Decision Tree	0.6641

Best Model

Property	Value
Model	Random Forest
Score	0.7449

Cross Validation

Metric	Value
Mean Score	0.7415
Standard Deviation	0.0071

Fold	Score
Fold 1	0.7436
Fold 2	0.742
Fold 3	0.7343
Fold 4	0.7343
Fold 5	0.7535

Hyperparameter Tuning

Parameter	Value
max_depth	20
min_samples_split	2
n_estimators	50

Best Validation Score: 0.7477

Feature Importance

Feature	Importance
num__model_year	0.0957
num__price	0.0957
num__milage	0.0837
cat__transmission_A/T	0.0183
cat__transmission_Automatic	0.0124
cat__int_col_Gray	0.01
cat__engine_305.0HP 3.7L V6 Cylinder Engine Gasoline Fuel	0.0076
cat__model_XTS Luxury	0.0074
cat__int_col_Beige	0.0074
cat__transmission_6-Speed A/T	0.0064

Model Recommendation

Property	Value
Recommended Model	Logistic Regression
Confidence	85

Recommendation Reasons

- Very stable across validation folds.
- Hyperparameter tuning improved performance.
- Simple and interpretable baseline model.
- Works well when classes are linearly separable.

Principal Component Analysis (PCA)

PCA analysis has not been performed.

Generated Visualizations

No visualization images were found.

Executive Summary

- Problem Type : classification
- Target Column : accident
- Recommended Model : Logistic Regression
- Confidence : 85
- Cross Validation Mean : 0.7415
- Cross Validation Std : 0.0071

Recommendations

- Use the recommended model for deployment.
- Validate predictions using unseen data before production use.
- Consider collecting additional data if model performance is unsatisfactory.
- Regularly retrain the model as new data becomes available.
- Monitor model performance over time.

End of Report

This report was automatically generated by DataMind AI.

Generated using FastAPI, Scikit-Learn, XGBoost and ReportLab.